LUHN ALGORITHM VALIDATOR

by

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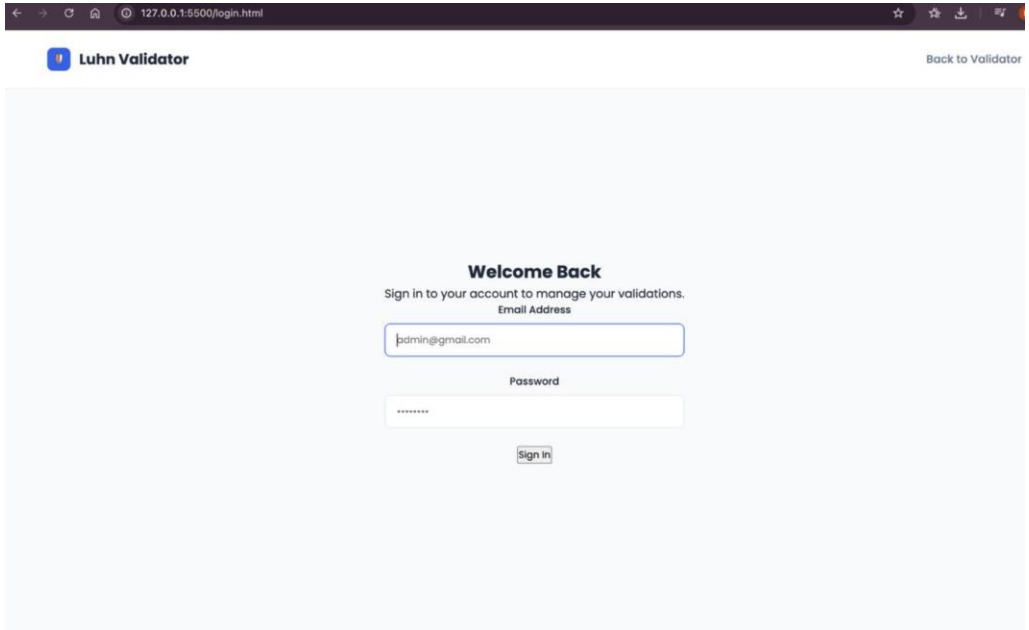
1. Problem Statement

In the modern digital world, online shopping and digital payments have become a regular part of daily life. People use credit cards and debit cards to make payments on websites and mobile applications. Because these transactions involve sensitive financial information, it is very important for systems to verify the details entered by users.

However, users may sometimes enter incorrect card numbers due to typing mistakes, missing digits, or incorrect formatting. If these errors are not detected at an early stage, they can result in failed transactions, payment errors, and poor user experience.

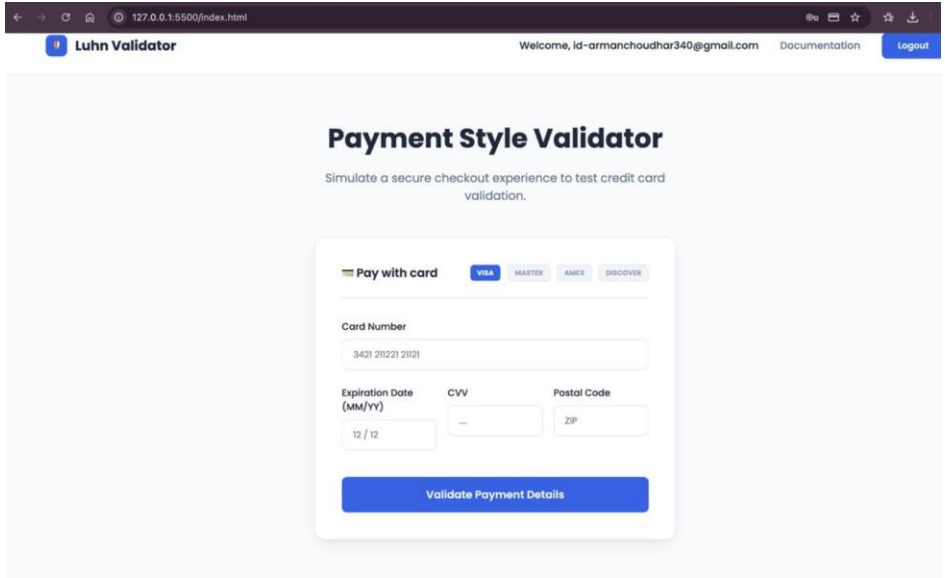
To solve this issue, payment systems use validation techniques that check whether the card number follows the correct structure before processing the transaction.

One of the most widely used methods is the Luhn Algorithm, developed by Hans Peter Luhn in 1954. This algorithm helps detect common errors in card numbers and is used by major payment networks such as Visa, MasterCard, American Express, and Discover as a preliminary validation step.



2. Solution

To address this problem, a web-based Luhn Algorithm Validator was developed using HTML, CSS, and JavaScript.
The system provides an interactive and user-friendly payment form interface where users can enter their card details. The form includes fields such as card number, expiry date, CVV, and postal code, similar to real online payment gateways.
When the user clicks the Validate Payment button, the system processes the input using JavaScript. The card number is checked using the Luhn Algorithm, while other fields are also validated to ensure they follow the correct format.
If all the details are valid, the system displays a success message confirming that the card number is valid. If any information is incorrect, the system shows clear error messages and highlights the incorrect fields so that the user can correct them.
This approach improves the accuracy, reliability, and usability of online payment systems.



3. Working/Methods & Algorithms.

The core functionality of the system is based on the Luhn Algorithm, which is a checksum formula used to validate identification numbers such as credit card numbers.

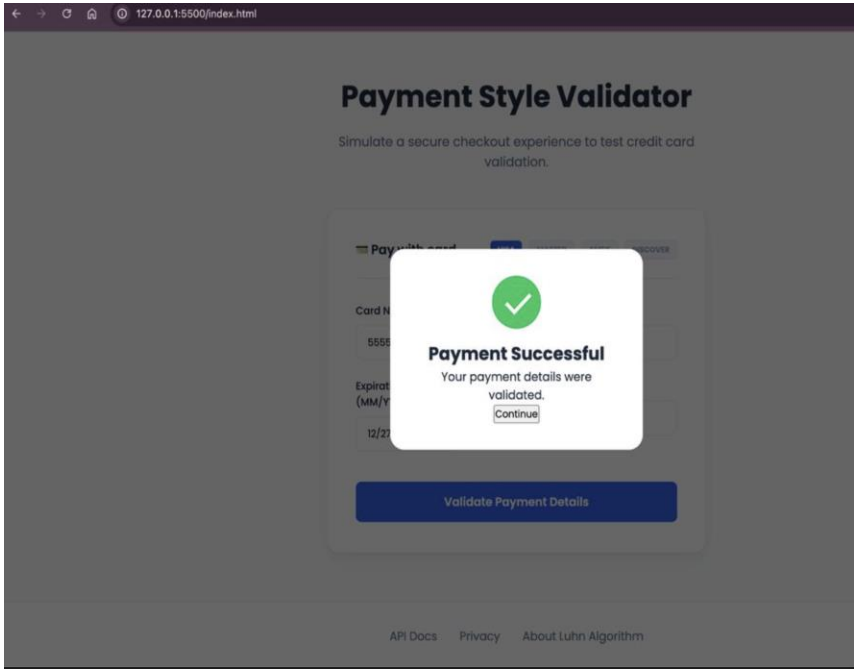
The algorithm works through the following steps:

- First, all spaces or non-numeric characters are removed from the card number.
- Starting from the rightmost digit, every second digit is doubled.
- If doubling a digit produces a number greater than 9, 9 is subtracted from it.
- After this, all digits are added together to calculate a total sum.
- If the final sum is divisible by 10, the card number is considered valid.

Along with the Luhn validation, the system also performs additional checks such as:

- Expiry date validation to ensure the card is not expired
- CVV validation using regular expressions
- Input formatting for better readability of card numbers
- Real-time feedback to guide the user

These methods ensure that incorrect or invalid inputs are detected before the transaction process continues.



4. Conclusion

The Luhn Algorithm Validator project successfully demonstrates how card number validation systems work in real-world payment applications.
By combining HTML, CSS, and JavaScript, the project creates an interactive web application that validates user input and provides immediate feedback. The system is capable of detecting incorrect card numbers, highlighting errors, and guiding users to enter valid information.
Through this project, several important concepts were explored, including algorithm implementation, form validation, DOM manipulation, and event handling in JavaScript.
Overall, this project provides a practical understanding of how modern payment systems perform basic card validation to prevent input errors and improve transaction reliability.
The system can be further enhanced by integrating backend verification, payment gateway APIs, and stronger security mechanisms for real-world deployment.
d file handling to manage a real-world parking system.

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Thank You